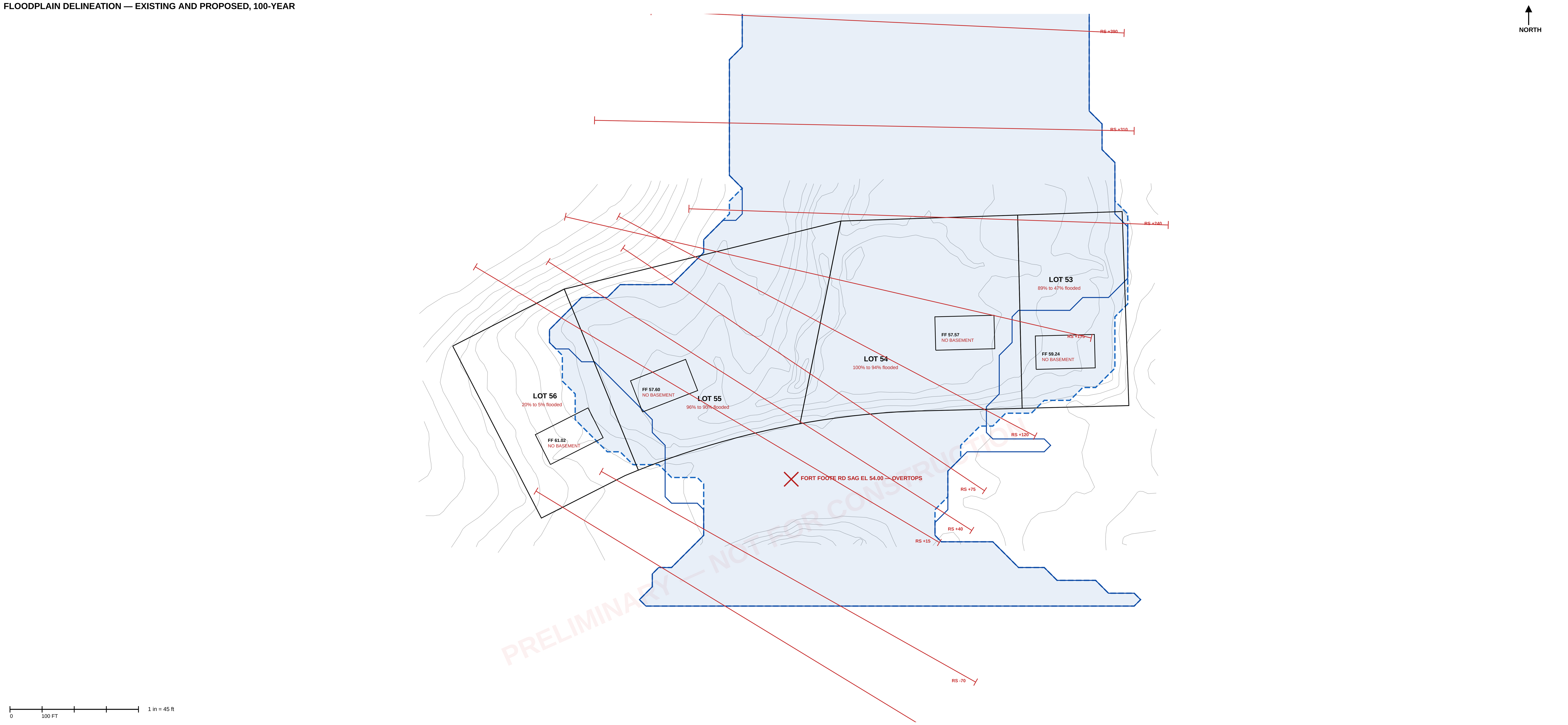
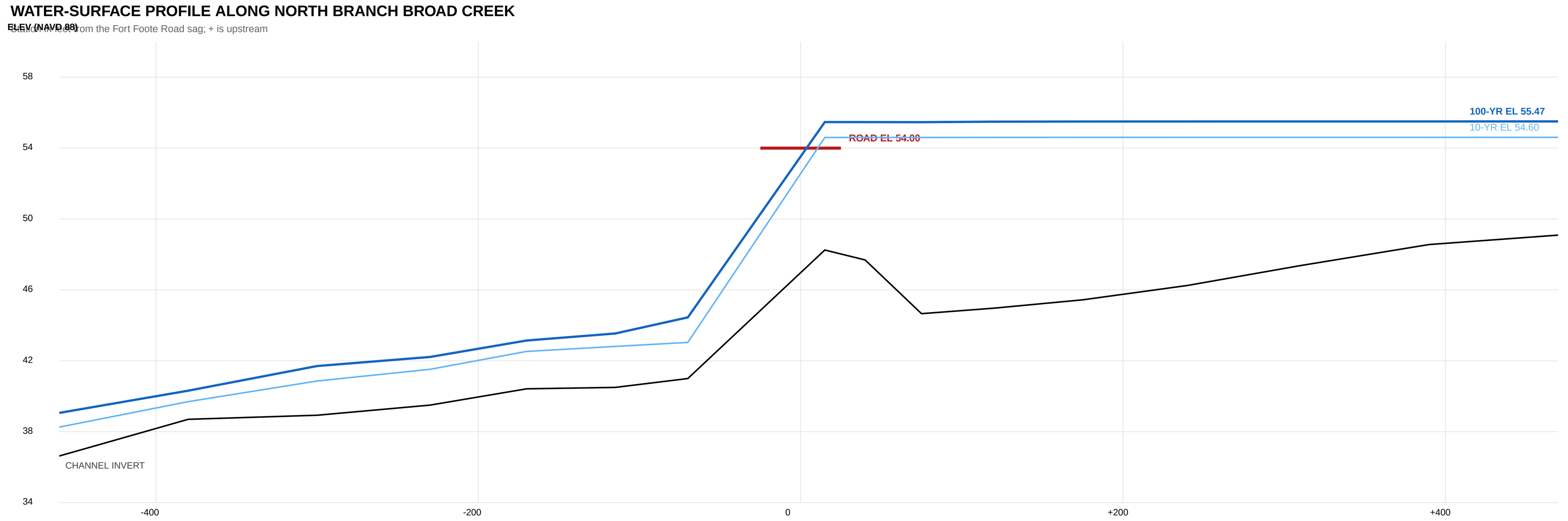
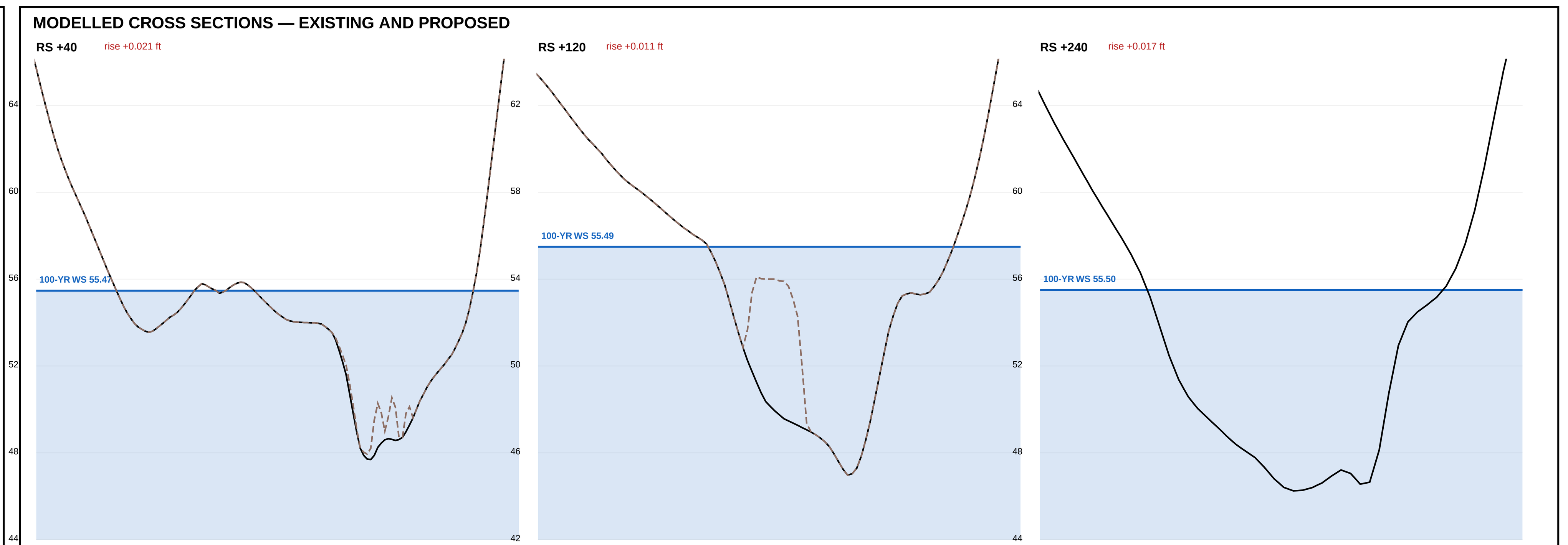
FLOODPLAIN DELINEATION — EXISTING AND PROPOSED, 100-YEAR



WATER-SURFACE PROFILE ALONG NORTH BRANCH BROAD CREEK



MODELLED CROSS SECTIONS — EXISTING AND PROPOSED



Solid = existing ground. Dashed brown = proposed grading. Shaded = 100-year water. Vertical exaggeration applies; stations in feet from the thalweg.

DESIGN DISCHARGES — NRCS TR-55, TYPE II

STORM	P24 (IN)	Q (IN)	qu	PEAK (CFS)
10-year	4.81	1.75	324	351
25-year	6.01	2.63	350	571
50-year	7.08	3.47	367	792
100-year	8.29	4.48	382	1,061
Area	397 ac	CN	68.0	Tc 0.86 hr
CULVERT AT Q100				
36 in	inlet	HW 55.45	HW/D 3.6	939 over rd
48 in	inlet	HW 55.36	HW/D 2.7	855 over rd
60 in	inlet	HW 55.26	HW/D 2.1	757 over rd
72 in	inlet	HW 55.13	HW/D 1.7	650 over rd
84 in	inlet	HW 55.00	HW/D 1.5	541 over rd
96 in	inlet	HW 54.87	HW/D 1.3	439 over rd

EXISTING vs PROPOSED WATER SURFACE — 100-YEAR

SECTION	EXISTING	PROPOSED	RISE (FT)	
RS +15	55.47	55.47	+0.000	
RS +40	55.47	55.49	+0.021	RISE
RS +75	55.46	55.49	+0.030	RISE
RS +120	55.49	55.50	+0.011	RISE
RS +175	55.50	55.51	+0.016	RISE
RS +240	55.50	55.52	+0.017	RISE
RS +310	55.50	55.52	+0.018	RISE
RS +390	55.50	55.52	+0.017	RISE
RS +470	55.51	55.52	+0.017	RISE
MAXIMUM			+0.030	NOT NO-RISE
Storage removed below the 100-yr surface			5,165 cy	
Total fill in the modelled reach			6,617 cy	
Compensatory storage			REQUIRED	

FLOODPLAIN ON THE LOTS, AND FREEBOARD — 100-YEAR

LOT	AREA (SF)	FLOODED EXIST	FLOODED PROP	FILL < FLOOD
LOT 53	12,500	11,100 (89%)	5,900 (47%)	771 cy
LOT 54	23,500	23,500 (100%)	22,100 (94%)	2,516 cy
LOT 55	25,600	24,500 (96%)	23,000 (90%)	1,678 cy
LOT 56	13,900	2,800 (20%)	700 (5%)	113 cy
LOWEST FLOOR CHECK AGAINST WS EL 55.52				
LOT	FIN FLOOR	FREEBOARD	BASEMENT	FOUNDATION
LOT 53	59.24	+3.72	none	slab
LOT 54	57.57	+2.05	none	slab
LOT 55	57.60	+2.08	none	slab
LOT 56	61.02	+5.50	none	slab

BASIS, METHOD AND LIMITATIONS

HYDROLOGY NRCS TR-55 graphical peak discharge, Type II distribution. Drainage area delineated by D8 routing on the USGS 3DEP bare-earth DEM. Curve number composited from the county 2023 impervious-surface and tree-canopy layers and the NRCS soil survey. Rainfall from NOAA Atlas 14 Vol. 2 Ver. 3, partial duration series, at 38.7611 N 77.0115 W.

HYDRAULICS One-dimensional steady gradually-varied flow, standard step, conveyance subdivided at the bank stations, average-conveyance friction slope. Crossing analysed to FHWA HDS-5 with both controls computed and the roadway treated as a broad-crested weir.

TERRAIN PGAtlas Contour 2 Ft (2023), NAVD 88, interpolated to a 10 ft grid. THIS IS NOT A FIELD SURVEY.

DATUM NAVD 88 = NGVD 29 - 0.66 ft here, from NGS marks HV4762, HV4763 and HV4728. On that basis the 57 ft WSSC of FPS-770017 is about EL 56.3 NAVD 88, against EL 55.47 computed. To be confirmed by levelling to benchmarks H31A, H32A and H32B on the field topographic survey.

NOT DONE No field survey. The existing culvert has never been measured and its size, inverts and entrance are assumed. HEC-RAS was not run and no HEC-RAS files exist. FPS 200546, the controlling study of record, has not been obtained. Upstream stormwater facilities were not inventoried. No floodway was delineated and none exists on current mapping. No unsteady or storage routing.

FLOODPLAIN OF RECORD FEMA FIRM panel 24033C0220E eff. 2016-09-16 maps these lots Zone X, outside the special flood hazard area, with no BFE and no floodway. That is a statement about FEMA mapping, not about flood risk. The county regulates this floodplain through the FPS series regardless.

EASEMENT The only platted floodplain easement in Indian Queen East (Plat 118-083, recorded 17 January 1984, 5.32 ac) lies entirely south of Fort Foote Road and covers none of Lots 53-56.

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PROJECT
Indian Queen East — Lots 53, 54, 55 and 56

ADDRESS
9588 · 9584 · 9580 · 9576 Fort Foote Road

JURISDICTION
Prince George's County, Maryland

ZONE
RSF-95

PLAT OF RECORD
Plat Book WWW 65, folio 60

SHEET
FP-101 — FLOODPLAIN STUDY AND DELINEATION, EXISTING AND PROPOSED

DISCIPLINE
Maryland Professional Engineer

HORIZONTAL DATUM
NAD 83, Maryland State Plane, US survey feet (EPSG:2248)

VERTICAL DATUM
NAVD 88

DATE
2026-09-11

STATUS

PRELIMINARY FEASIBILITY. NOT A SEALED FLOODPLAIN STUDY. No field survey. Culvert never measured. HEC-RAS not run. FPS 200546, the controlling study of record, not obtained. No professional engineer has reviewed or sealed this sheet.

PROFESSIONAL CERTIFICATION

Maryland P.E. No. ________ Date __________

KEY FINDINGS

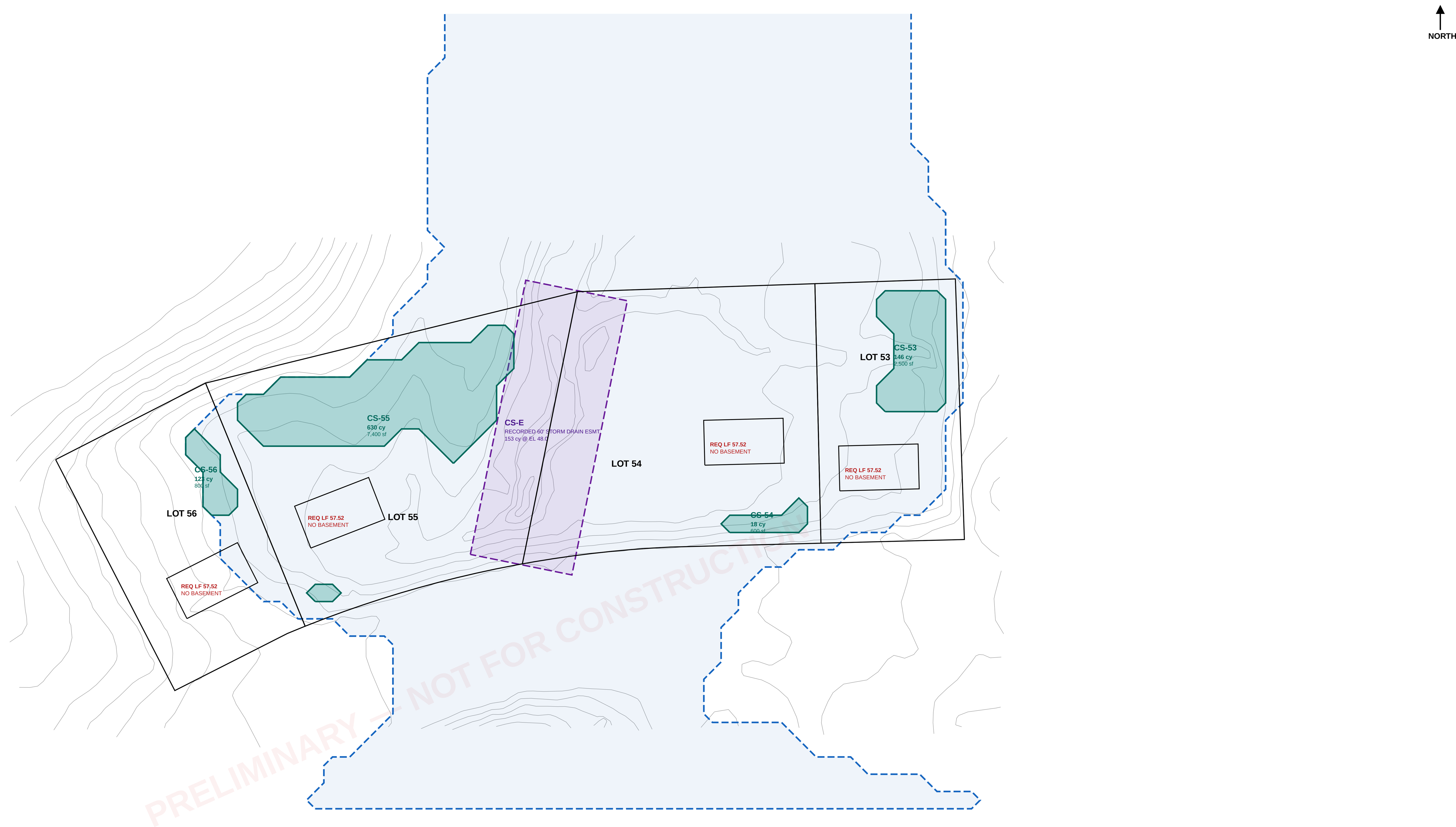
Contributing drainage area	**397 ac (0.620 sq mi)**
Composite curve number	**68.0**
Time of concentration	**0.86 hr**
Q100 at the crossing	**1,061 cfs**
Fort Foote Road sag	**EL 54.00**
100-yr water surface, existing	**EL 55.47**
100-yr water surface, proposed	**EL 55.52**
Max rise, proposed vs existing	**+0.03 ft — NOT no-rise**
Flood storage removed by fill	**5,165 cy**
Road overtops at Q100	**956 cfs over the pavement**

LEGEND

- 100-yr floodplain limit, EXISTING
- 100-yr floodplain limit, PROPOSED
- Modelled cross section
- Lot boundary of record
- Existing contour (2 ft, county LiDAR)

COMPENSATORY STORAGE DESIGN — PLAN

Cells CS-53 to CS-56 excavated to EL 50.50; cell CS-E within the recorded 60 ft storm drain easement



0 100 FT 1 in = 42 ft

RESERVOIR ROUTING — EXISTING CROSSING

STORM	INFLOW	OUTFLOW	ATTENUATED	PEAK STAGE
10-year	388 cfs	215 cfs	44.6%	EL 54.31
25-year	606 cfs	512 cfs	15.5%	EL 54.82
50-year	814 cfs	758 cfs	6.8%	EL 55.13
100-year	1,058 cfs	1,032 cfs	2.4%	EL 55.43

CROSSING ALTERNATIVES — DOWNSTREAM TRANSFER

OPTION	10-YR OUT	CHANGE	100-YR OUT	CHANGE
existing 36 in	215	+0	1,032	+0
48 in RCP	179	-36	1,023	-9
72 in RCP	275	+60	981	-52
twin 72 in RCP	333	+118	870	-162
8 x 6 ft box	307	+91	953	-80
10 x 8 ft box	347	+132	869	-163
twin 10 x 8 ft box	388	+173	974	-58
20 x 8 ft box (bridge-class)	388	+173	974	-58
twin 20 x 8 ft box	388	+173	1,058	+26

COMPENSATORY STORAGE SCHEDULE

CELL	LOT	AREA (SF)	MEAN DEPTH	VOLUME
CS-53	53	2,500	1.57 ft	146 cy
CS-54	54	600	0.79 ft	18 cy
CS-55	55	7,400	2.30 ft	630 cy
CS-56	56	800	4.16 ft	123 cy
CS-E	54/55	9,721	to EL 48.0	153 cy

PROVIDED	1,070 cy
REQUIRED by proposed grading	5,165 cy
DEFICIT	4,095 cy

FILL BELOW EL 55.53 — DISTRIBUTION

beneath dwelling footprints	1,985 cy
within 10 ft of a dwelling	1,410 cy
driveways, aprons, tie-out	1,770 cy

RESOLUTION: reduce fill. Vented stem-wall or pier foundations and driveways at existing grade bring the balance within the volume provided.

FLOODPLAIN REMOVAL AND FOUNDATION DETERMINATION

LOT	LOWEST GROUND	WS MUST FALL BELOW	REQUIRED DROP	ACHIEVABLE
LOT 53	EL 48.78	EL 48.78	6.69 ft	partial
LOT 54	EL 44.97	EL 44.97	10.50 ft	partial
LOT 55	EL 44.66	EL 44.66	10.81 ft	partial
LOT 56	EL 53.34	EL 53.34	2.13 ft	partial

Controlling tailwater lower bound: EL 44.45
The upstream water surface cannot be lowered below the downstream water surface.

LOWEST FLOOR DETERMINATION

LOT	PROPOSED FF	BASEMENT	REQ LOWEST FLOOR	STATUS
LOT 53	59.24	none	57.52	COMPLIES
LOT 54	57.57	none	57.52	COMPLIES
LOT 55	57.60	none	57.52	COMPLIES
LOT 56	61.02	none	57.52	COMPLIES

Basements are not permissible on any lot.

FREQUENCY OF INUNDATION — EXISTING GROUND

STORM	STAGE	LOT 54	LOT 55	ROAD
2-yr	EL 50.17	89%	41%	clear
5-yr	EL 53.02	97%	78%	clear
10-yr	EL 54.31	99%	92%	OVERTOPPED
25-yr	EL 54.82	100%	93%	OVERTOPPED
50-yr	EL 55.13	100%	95%	OVERTOPPED
100-yr	EL 55.43	100%	95%	OVERTOPPED

RECOMMENDED: DEVELOP LOTS 53 AND 56;
DEDICATE LOTS 54 AND 55 AS FLOODPLAIN

compensation needed by 53 + 56	891 cy
available on 54 + 55 to EL 50.0	1,399 cy
ratio provided to required	1.6 : 1

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PROJECT
Indian Queen East — Lots 53, 54, 55 and 56

ADDRESS
9588 · 9584 · 9580 · 9576 Fort Foote Road

JURISDICTION
Prince George's County, Maryland

SHEET
FP-102 — MITIGATION ANALYSIS AND COMPENSATORY STORAGE DESIGN

DISCIPLINE
Maryland Professional Engineer

VERTICAL DATUM
NAVD 88

DATE
2026-09-11

STATUS
PRELIMINARY FEASIBILITY. NOT A SEALED FLOODPLAIN STUDY. Quantities are for concept evaluation and are not construction quantities. No professional engineer has reviewed or sealed this sheet.

PROFESSIONAL CERTIFICATION

Maryland P.E. No. ________ Date __________

DETERMINATIONS

RECOMMENDED: develop 53 and 56	**DEDICATE 54 AND 55**
Removal of all lots from floodplain	**NOT ACHIEVABLE**
Controlling tailwater, lower bound	**EL 44.45**
Lowest ground, Lot 55	**EL 44.66**
Basements	**NOT PERMISSIBLE, ALL LOTS**
Required lowest floor	**EL 57.52**
Compensation required	**5,165 cy**
Compensation available on site	**1,070 cy**
Crossing enlargement	**NOT RECOMMENDED**
Lot 54 flooded, 2-year event	**89%**
Road overtopped	**BETWEEN 5- AND 10-YEAR**
Storage to dry the rear yards	**136,670 cy — NOT FEASIBLE**
Upstream diversion benefit	**0.07 ft**
53 + 56 only: compensation needed	**891 cy**
available on 54 + 55 at EL 50	**1,399 cy**

DESIGN CRITERIA — COMPENSATORY STORAGE

1 Excavation floor EL 50.50, above the invert of the receiving channel, so each cell drains by gravity.
2 Cells located within the existing-condition 100-year floodplain limit, being the area to be placed under floodplain easement.
3 Side slopes no steeper than 3:1, stabilised and mowable, no retaining structure.
4 Minimum 10 ft from every property line and 20 ft from every structure.
5 Positive hydraulic connection to the recorded storm drain easement corridor at the cell invert.
6 Volume provided at the same elevation increments from which it is taken; stage-by-stage comparison required at final design.